

Assignment IV

DATA301/DATA471

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General instructions

You may submit your work as a PDF. However, we **strongly recommend** submitting it as a Quarto or R Markdown document.

If you submit a Quarto or R Markdown document, we **prefer that you provide a link to a Git repository** containing all the files needed to reproduce your work. We will clone the repository and render the document ourselves.

Exercise 1

Given an inference problem, a good practice is to use more than one estimator. The estimators you choose should have different properties to cover a variety of situations. We usually choose among:

- maximum likelihood estimators: they have “the best” properties when the model is correct for the data,
- estimators based on analogy: they are usually simple to compute,
- robust estimators: they are not too bad when the model is correct, but still perform well under departures from the hypothesis, e.g., in the presence of surprising observations.

Imagine that you are studying the time between successive arrivals of customers at a service desk. You collect data, the times between arrivals, and decide that an exponential distribution is a reasonable model for these observations.

The exponential distribution is particularly useful in situations like this

You have samples of independent identically distributed random variables $\mathbf{X} = (X_1, X_2, \dots, X_n)$ with common Exponential distribution with parameter $\lambda > 0$, the reciprocal of the mean. You have two estimators for λ obtained by maximum likelihood $\hat{\lambda}_{\text{ML}}$ and by analogy with the inter-quartile range $\hat{\lambda}_{\text{IQR}}$:

$$\hat{\lambda}_{\text{ML}}(\mathbf{X}) = \left(\frac{1}{n} \sum_{i=1}^n X_i \right)^{-1}, \text{ and} \quad (1)$$

$$\hat{\lambda}_{\text{IQR}}(\mathbf{X}) = \frac{\ln 3}{q_{3/4}(\mathbf{X}) - q_{1/4}(\mathbf{X})}, \quad (2)$$

where $q_{3/4}(\mathbf{X})$ is the third quartile of $\mathbf{X}$, and $q_{1/4}(\mathbf{X})$ is the first quartile of $\mathbf{X}$.

These estimators use the data in rather different ways. The maximum likelihood estimator uses the entire sample through the likelihood, while the IQR estimator focuses on a single feature of the distribution: the difference between the third and first quartiles.

[10%] There are several ways of computing order statistics, e.g., the quartiles that you need for $\hat{\lambda}_{\text{IQR}}(\mathbf{X})$. Read and summarise the analysis of alternatives made by Wicklin (2017). This article gives you a link to the article that sparked the discussion (Hyndman & Fan, 1996), and we strongly recommend you also read it. Which type of order statistics would you use and why? Which is the default type of order statistics your version of R/Python uses?

[20%] You suspect that $\hat{\lambda}_{\text{IQR}}$ is biased, so derive and implement a first-order bootstrap bias corrected estimator based on $\hat{\lambda}_{\text{IQR}}$. Denote it $\tilde{\lambda}$. You may fix the number of bootstrap replications in $B = 200$ for every input sample.

[20%] Make a Monte Carlo experiment comparing these three estimators. Use as parameter space $\Lambda \times N$, where $\lambda \in \Lambda = \{.1, 1, 10\}$ and $n \in N = \{5, 30, 100\}$. Use the mean square error as the sole metric for your analysis. Report your results. Justify your choice(s) of the number of replications for each λ and each n . You may want to check the book by Robert & Casella (2009); it is available in our library.

Exercise 2

Consider the `ausbeer` data from the `fpp2` R package: quarterly beer production in Australia. This dataset is thoroughly analysed in the wonderful book by Hyndman & Athanasopoulos (2018). You should check what the authors did!

[50%] Produce one or more visualisations and answer:

1. What is the frequency of the series? How can you determine this from the data and/or the time-series plot?
2. Describe the overall behaviour of the series. Is there evidence of a long-term trend?
3. Is there evidence of seasonality? If so, describe the seasonal pattern.
4. Does the variability appear constant over time? Explain what you see in the plot and whether a transformation might be appropriate.
5. Are there any observations that you would regard as unusual? Identify them and explain why they attract attention.
6. Plot the series by season (for example, using a seasonal plot). What does this reveal that is less obvious in the ordinary time-series plot?
7. Plot the seasonal subseries. Which quarters have the highest and lowest typical values? Is the seasonal pattern stable over time?
8. Would you detrend or transform the series before modelling it? Give a visual argument for your answer. Do not fit a model.

References

- Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting: Principles and practice*. OTexts. [OTexts.com/fpp2](https://otexts.com/fpp2)
- Hyndman, R. J., & Fan, Y. (1996). Sample quantiles in statistical packages. *The American Statistician*, 50(4), 361. <https://doi.org/10.2307/2684934>
- Robert, C. P., & Casella, G. (2009). *Introducing Monte Carlo methods with R*. Springer.
- Wicklin, R. (2017). *Sample quantiles: A comparison of 9 definitions*. SAS Blogs. <https://blogs.sas.com/content/iml/2017/05/24/definitions-sample-quantiles.html>